

Experiment 1: Error in Experimental Data

Objectives:

No physical quantity can be measured with perfect certainty. Therefore, in this experiment, we aim to learn some basic principles of error analysis.

The experiment is divided into two parts. Using two resistor samples, part A aims to introduce the basis of statistical sampling analysis for the error in resistors due to their tolerance. Then, part B introduces an additional source of error in the electrical circuits, besides the components' tolerance, which is the insertion error of measurements instrumentation.

Components: twenty 100 K Ω resistors, AVO meter, one 1 K Ω resistor, test board, and wires. (*Note if you can't find 100 K Ω resistor, bring any other resistance value*)

Part A: Statistical analysis for resistors' samples:

First, we start with a 10-resistors sample—sample 1.

1. Record the true value (R_t) of the resistor patch from the color code.
2. Measure and record the resistance of each resistor.
3. Calculate the average resistance ($\bar{R}$) of the first ten samples.

$$\bar{R} = \frac{\sum_{i=1}^n R_i}{n}$$

1. For each resistor, calculate the percentage error and deviation. Such that for a resistor number i (R_i) will have its percentage error ($\%e_i$) and its deviation (d_i).

$$d_i = R_i - \bar{R}$$

$$\%e_i = \frac{|R_t - R_i|}{R_t} * 100$$

2. Calculate the standard deviation of the sample (σ)

$$\sigma = \sqrt{\frac{\sum_{i=1}^n d_i^2}{n}}$$

Then, we repeat from 1 to 5 for a 20-resistors sample—sample 2.

Resistor no.	R_i (K Ω)	d_i (K Ω)	d_i^2	%e
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
$\bar{R} = \frac{\sum_{i=1}^{10} R_i}{10} = \text{-----}$ $\sigma = \sqrt{\frac{\sum_{i=1}^{10} d_i^2}{10}} = \text{-----}$				

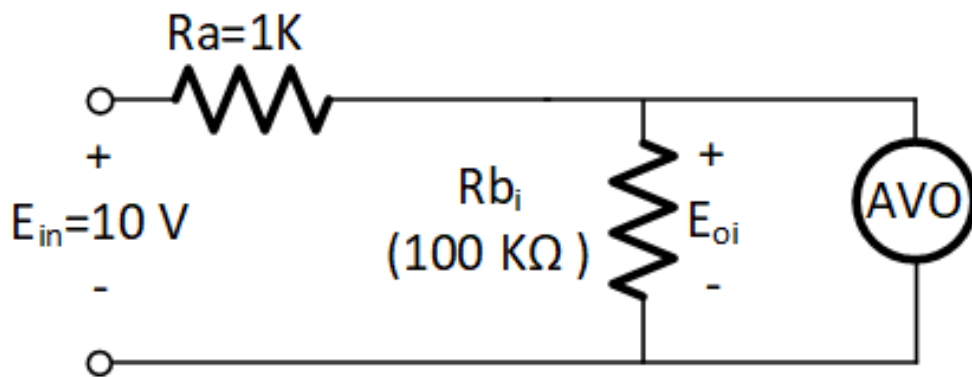
Resistor no.	R_i (K Ω)	d_i (K Ω)	d_i^2	%e
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
$\bar{R} = \frac{\sum_{i=1}^{20} R_i}{20} = \text{-----}$ $\sigma = \sqrt{\frac{\sum_{i=1}^{20} d_i^2}{20}} = \text{-----}$				

Question: What is the distribution curve name for the resistor readings?
Draw it.

Question: Do the results from sample 1 or sample 2 will be able to construct the distribution curve accurately? and why?

Part B: Insertion Error

1. Set the following circuit where $E_{in}=10\text{ V}$, $R_a= 1\text{K}\Omega$, and R_b is to be selected from the $100\text{k}\Omega$ resistors. We will select 5 resistors for R_b each one is called R_{b1} , R_{b2} , R_{b3} , R_{b5}
2. For each R_b of the five resistors, measure the actual resistance value (R_{bi}), record this value.
3. For each R_{bi} value, use it to calculate E_{oi} (true) using circuit analysis record them; Then, use The AVO meter to measure the output voltage E_{oi} (measured).
4. Record E_{oi} (true) and E_{oi} (measured) values in the following table.



5. Calculate the insertion error for each $100\text{k}\Omega$ resistor

$$\% \text{insertion error for } R_{bi} (\%e_i) = \frac{|E_{oi}(\text{true}) - E_o(\text{measured})|}{E_{oi}(\text{true})} * 100$$

Resistor no. (i)	R_{bi} Measured	E_{oi} True	E_{oi} Measured	$\%e_i$
1				
2				
3				
4				
5				

Question: What is meant by the insertion error?

Question: what will happen to the insertion error if we replaced the digital AVO meter with an analog meter?
